

Concept Revision Notes

Everything you need to know to solve the questions in **School Paper Based** — the Heritage School's two Mathematics papers and the similar sums that go with them — arranged chapter by chapter.

Nothing here is general revision. Every item is on the page because one or more of those questions needs it, and each says which. Where a concept has a standard way of going wrong, that is set out too.

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1. Relations and Functions

Four questions, and between them they test every definition in the chapter. Nothing here needs calculation — it is all about stating a definition precisely and then checking it.

1.1 The three properties, with their quantifiers

Reflexive: $(a, a) \in R$ for **every** a in the set. **Symmetric:** whenever $(a, b) \in R$, also $(b, a) \in R$. **Transitive:** whenever (a, b) and (b, c) are both in R , so is (a, c) .

WATCH OUT Reflexive needs *every* element, so one missing pair kills it. Symmetric and transitive are conditional — they only ask for something when the premise holds.

NEEDED FOR FT Q1(xv), FT Q8

1.2 Vacuous truth

If a relation contains no pair (a, b) and (b, c) to trigger transitivity, the condition is satisfied **vacuously** and the relation *is* transitive.

WATCH OUT This is the commonest one-mark trap in the chapter. $R = \{(1, 2)\}$ is transitive; $R = \{(2, 2)\}$ is symmetric *and* transitive.

NEEDED FOR FT Q1(xv), CISCE competency Q28

1.3 Equivalence relation and its classes

All three properties together. The classes partition the set, and two elements are in the same class exactly when they are related.

WATCH OUT A parity condition is transitive; a **bounded-difference** condition such as $|a - b| \leq 5$ is not — differences add but the bound does not.

NEEDED FOR FT Q8, NCERT Ch 1 ($|a - b|$ even)

1.4 One-one, onto, bijection — how each is proved

One-one: assume $f(x_1) = f(x_2)$ and derive $x_1 = x_2$. **Onto:** take any y in the co-domain and solve $y = f(x)$ for an x in the domain.

NEEDED FOR FT Q1(vi), Exemplar Ch 1 Q21(iii)

1.5 Strictly increasing $\Rightarrow$ one-one

If f is strictly increasing (or strictly decreasing) on an interval it is automatically one-one there. With continuity, the intermediate value theorem then makes it onto its range.

WATCH OUT $x|x|$ is strictly increasing across 0 because the modulus keeps the sign; x^2 is not.

NEEDED FOR FT Q1(vi)

1.6 Range against co-domain

The co-domain is where the function is *declared* to land; the range is what it actually reaches. A question asking for the range wants the second.

NEEDED FOR FT Q1(ii), Exemplar Ch 1 Q14

1.7 Finding a range by completing the square

$f(x) = x^2 - 4x + 5 = (x - 2)^2 + 1$. On a restricted domain, read off what the bracket can do and push it through.

WATCH OUT The restriction to $[2, \infty)$ is what makes the function one-one; without it the same formula is many-one.

NEEDED FOR FT Q1(ii)

2. Inverse Trigonometric Functions

Five questions. Every one of them turns on the **principal value branch** — the algebra is easy, and the marks are in knowing when an identity is allowed.

2.1 The principal value branches

FUNCTION	BRANCH
$\sin^{-1} x$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$
$\cos^{-1} x$	$[0, \pi]$
$\tan^{-1} x$	$(-\frac{\pi}{2}, \frac{\pi}{2})$
$\cot^{-1} x$	$(0, \pi)$

WATCH OUT Nothing else in this chapter matters as much. Learn the four.

NEEDED FOR FT Q1(iii), FT Q7, PB Q3(b), Exemplar Ch 2 Q5

2.2 The two complementary identities

$$\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}, \quad \tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$$

WATCH OUT Whenever two different inverse functions of the *same* x appear, reach for these first — they usually collapse the whole expression.

NEEDED FOR FT Q1(iii), FT Q3, Exemplar Ch 2 Q22, Exemplar Ch 2 Q26

2.3 Folding an angle back into its branch

$f^{-1}(f(\theta)) = \theta$ **only** when θ is in the branch. Outside it, subtract the right period: π for $\tan^{-1}$, 2π for $\sin^{-1}$ and $\cos^{-1}$.

WATCH OUT Using 2π on a tangent is the usual error. The answer to FT Q7 is $\frac{9\pi}{4} - x$, *not* $\frac{\pi}{4} - x$, purely because of the interval given.

NEEDED FOR FT Q7, Exemplar Ch 2 Q5

2.4 Converting one inverse function into another

Draw the right triangle. $\sin^{-1}\frac{4}{5}$ has opposite 4 and hypotenuse 5, so adjacent 3 and the angle is also $\tan^{-1}\frac{4}{3}$.

WATCH OUT Only valid for the acute case; check the sign of the cosine before assuming it.

NEEDED FOR PB Q1(b), Exemplar Ch 2 Q14

2.5 The $\tan^{-1}$ addition formula, and its condition

$$\tan^{-1}x + \tan^{-1}y = \tan^{-1} \frac{x+y}{1-xy}, \quad xy < 1$$

WATCH OUT When $xy > 1$ a $\pm\pi$ must be added — that is what the condition is protecting you from.

NEEDED FOR PB Q1(b), NCERT Ch 2

2.6 Naming the angles, then expanding

For $\sin^{-1}A \pm \sin^{-1}B$, set $A = \sin\alpha$, $B = \sin\beta$, read $\cos\alpha$ and $\cos\beta$ off the triangles, and expand $\sin(\alpha \pm \beta)$.

WATCH OUT Confirm the sign of $\cos(\alpha \pm \beta)$ before quoting the answer as an inverse sine — the obtuse angle has the same sine.

NEEDED FOR PB Q3(b), Exemplar Ch 2 Q14

2.7 Compressing a $\sin x + b \cos x$

$$\frac{\sin x + \cos x}{\sqrt{2}} = \sin\left(x + \frac{\pi}{4}\right) = \cos\left(\frac{\pi}{4} - x\right). \text{ In general } a \sin x + b \cos x = \sqrt{a^2 + b^2} \sin(x + \phi).$$

NEEDED FOR FT Q7, Exemplar Ch 5 Q37

2.8 Test every candidate root

Squaring or cross-multiplying inside an inverse-trig equation can manufacture roots that the branches then reject.

WATCH OUT In PB Q3(b) the algebra gives $x = 2$ and $x = -1$; only $x = 2$ survives.

NEEDED FOR PB Q3(b), Exemplar Ch 2 Q7

3. Matrices

Four questions: one on transposes, one decomposition, and two systems of equations solved by the inverse.

3.1 Transpose rules

$(A')' = A$, $(A + B)' = A' + B'$, and — the one that matters — $(AB)' = B'A'$, with the order reversed.

WATCH OUT Keeping the order gives the wrong sign and therefore the wrong answer.

NEEDED FOR FT Q1(i), Exemplar Ch 3 Q63

3.2 Symmetric and skew symmetric

$A' = A$ is symmetric; $A' = -A$ is skew. A skew matrix always has a **zero diagonal**, because $a_{ii} = -a_{ii}$.

WATCH OUT The zero diagonal is the quickest check that an answer is right.

NEEDED FOR FT Q1(i), PB Q1(a)

3.3 The unique split

$$A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A')$$

WATCH OUT First bracket symmetric, second skew; the decomposition exists for every square matrix and is unique.

NEEDED FOR PB Q1(a), NCERT Ch 3 Ex 3.3, Exemplar Ch 3 Q52

3.4 Multiplication does not commute

$AB \neq BA$ in general, which is why $(A + B)^2 = A^2 + 2AB + B^2$ holds only when $AB = BA$.

NEEDED FOR FT Q1(i)

3.5 Inverse by the adjoint

$$A^{-1} = \frac{1}{|A|} \text{adj } A, \quad |A| \neq 0$$

WATCH OUT **adj A is the transpose of the cofactor matrix.** Forgetting the transpose is the single commonest way to lose these marks.

NEEDED FOR PB Q2(b), FT Q11

3.6 Solving a system by the matrix method

Write $AX = B$, check $|A| \neq 0$, then $X = A^{-1}B$. A unique solution exists precisely when $|A| \neq 0$.

WATCH OUT Always substitute the answer back into all three equations — it costs ten seconds and catches every arithmetic slip.

NEEDED FOR FT Q11, PB Q2(b), Exemplar Ch 4 Q19, NCERT Ch 4

3.7 Turning a word problem into a system

One variable per unknown price or quantity, one equation per purchase or condition. The matrix work is then mechanical.

NEEDED FOR FT Q11, NCERT Ch 4 (onion, wheat and rice)

4. Determinants

Four questions, none of which should be expanded by brute force. Every one is a properties question.

4.1 The properties worth knowing by name

- A common factor in a row or column comes out.
- Two identical — or proportional — rows or columns make it zero.
- $R_i \rightarrow R_i + kR_j$ leaves the value unchanged.
- Interchanging two rows changes the sign.

NEEDED FOR FT Q1(iv), FT Q6, PB Q2(a), Exemplar Ch 4 Q6, Q7, Q13

4.2 The standard opening move

$R_1 \rightarrow R_1 + R_2 + R_3$ (or the same with columns) often makes a whole row constant. Take that factor out, then clear the row with two more operations.

WATCH OUT Two operations are usually needed after the factor comes out; stopping after one leaves a determinant you still have to expand.

NEEDED FOR PB Q2(a), FT Q1(iv), Exemplar Ch 4 Q13

4.3 Determinants of products and powers

$$|AB| = |A||B|, \quad |A^n| = |A|^n, \quad |kA| = k^n|A| \text{ for an } n \times n \text{ matrix}$$

WATCH OUT $|kA| = k^n|A|$: the exponent is the **order**, not 1.

NEEDED FOR FT Q4

4.4 Adjoint identities

$$A(\text{adj } A) = |A| I_n, \quad |\text{adj } A| = |A|^{n-1}$$

NEEDED FOR FT Q4 (and the family it belongs to)

4.5 Checking a determinant identity cheaply

Substitute convenient numbers — $a = b = c = 1$, or $x = 0$ — into both sides. It will not prove the identity but it catches a wrong constant at once.

NEEDED FOR FT Q1(iv), PB Q2(a)

4.6 Spotting a linear relation between columns

If one column is a combination of the others the determinant is zero. Look for it before expanding: $(p + q)^2 - (p - q)^2 = 4pq$ is what makes FT Q6 vanish.

NEEDED FOR FT Q6, Exemplar Ch 4 Q7

5. Continuity and Differentiability

The heaviest chapter in both papers — eight questions. Half of them are ‘prove this second-order identity’, which is one template applied four times.

5.1 Continuity at a point

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$$

WATCH OUT All three parts, not two: the limits can agree with each other and still differ from the value.

NEEDED FOR FT Q1(ix), FT Q12, Exemplar Ch 5 Q11, Q16

5.2 Differentiability implies continuity

A function differentiable at a point is continuous there. The converse is false — $|x|$ at 0 is the standard counter-example.

WATCH OUT So a ‘derivable’ question carries **two** conditions: match the values and match the one-sided slopes.

NEEDED FOR FT Q12, CISCE competency Q22

5.3 Piecewise functions: two conditions

At the join, continuity gives one equation and equality of the one-sided derivatives gives another. Two unknowns need both.

WATCH OUT Matching only the slopes leaves the constant free and the function able to jump.

NEEDED FOR FT Q12, Exemplar Ch 5 Q16

5.4 Parametric first derivative

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

NEEDED FOR FT Q1(vii), FT Q1(viii), Exemplar Ch 6 Q44

5.5 Parametric second derivative — the trap

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}}$$

WATCH OUT It is **not** $\frac{d^2y}{dt^2} \div \frac{d^2x}{dt^2}$. FT Q1(viii) is built so that the wrong rule gives the assertion's wrong answer — both halves of it are false.

NEEDED FOR FT Q1(viii)

5.6 Simplify before you differentiate

$e^{\log t} = t$ for $t > 0$, so $y = e^{\log \sin^{-1} x} + e^{\log \cos^{-1} x}$ is just $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$.

WATCH OUT A constant differentiates to 0. Grinding through the chain rule gets there too, but slowly and with two chances to slip.

NEEDED FOR FT Q2, Exemplar Ch 5 Q37

5.7 Logarithmic differentiation

Take logs of both sides first when the unknown appears in an exponent or as a product of many factors.

WATCH OUT $(ax + b)e^{\frac{y}{x}} = x$ becomes $y = x \log x - x \log(ax + b)$ in one line.

NEEDED FOR FT Q9

5.8 The second-order identity template

1. Differentiate once and rearrange so the awkward root sits alone.
2. Square both sides to clear it.
3. Differentiate again.
4. Divide through by $2 \frac{dy}{dx}$, noting it is not identically zero.

WATCH OUT This one template answers every 'show that $(1 \pm x^2)y'' \pm xy' = \dots$ ' question in either paper.

NEEDED FOR FT Q9(OR), PB Q5(a), Exemplar Ch 5 Q81

5.9 Standard limits and small-angle expansions

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \sin x = x - \frac{x^3}{6} + \dots, \quad \cos x = 1 - \frac{x^2}{2} + \dots$$

WATCH OUT Two terms of each series settle a $\frac{0}{0}$ limit faster than three applications of L'Hôpital.

NEEDED FOR PB Q1(d)

5.10 Derivatives of the inverse trigonometric functions

$$\frac{d}{dx} \sin^{-1}x = \frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx} \tan^{-1}x = \frac{1}{1+x^2}$$

NEEDED FOR FT Q2, Exemplar Ch 5 Q37

6. Applications of Derivatives

Eight questions: two rates, three tangent-and-normal, three optimisation. The optimisation ones all have a constraint that must be used to remove a variable.

6.1 Related rates

Differentiate the relation with respect to **t**, then substitute the instantaneous values. The rate asked for is the one on the left; everything else is data.

NEEDED FOR Exemplar Ch 6 Q35, Q3 (the kite)

6.2 A rate with respect to something other than time

FT Q14 asks for $\frac{dV}{dh}$, not $\frac{dV}{dt}$. Express V in terms of that one variable and differentiate once.

WATCH OUT No $\frac{dh}{dt}$ is given and none is needed — read which variable the question actually names.

NEEDED FOR FT Q14

6.3 Increasing and decreasing from the sign of f'

$f' > 0$ on an interval $\Rightarrow$ strictly increasing there. Factorise f' and read the sign off the roots.

WATCH OUT A positive-leading quadratic is negative **between** its roots; a negative-leading one is positive between them.

NEEDED FOR FT Q1(x), Exemplar Ch 6 Q46

6.4 Tangent and normal

Slope of the tangent is $\frac{dy}{dx}$ at the point; the normal's slope is $-1 \div$ that. Then use $y - y_1 = m(x - x_1)$.

WATCH OUT $-\frac{1}{m}$, not $-m$.

NEEDED FOR FT Q5, FT Q5(OR), Exemplar Ch 6 Q41, Q14

6.5 'The tangent is this particular line'

Matching the slope is **necessary but not sufficient**. Find every point with that slope, then check which of them actually lies on the given line.

WATCH OUT In FT Q5 two points have slope 1 and only one lies on $y = x - 11$.

NEEDED FOR FT Q5

6.6 Implicit differentiation

Differentiate both sides as they stand, treating y as a function of x , then solve for $\frac{dy}{dx}$.

WATCH OUT $\sqrt{x} + \sqrt{y} = 4$ gives $\frac{dy}{dx} = -\frac{\sqrt{y}}{\sqrt{x}}$ — always negative, which decides the sign when the tangent is ‘equally inclined’.

NEEDED FOR FT Q5(OR), Exemplar Ch 6 Q14

6.7 Maxima and minima: the two tests

First derivative: f' changes $+$ to $-$ at a maximum. **Second derivative:** $f' = 0$ and $f'' < 0$ gives a maximum.

WATCH OUT If f'' is also zero the second test says nothing — fall back on the sign of f' .

NEEDED FOR PB Q5(b), FT Q14(OR), Exemplar Ch 6 Q31

6.8 On a closed interval, test the endpoints

The absolute maximum of a continuous function on $[a, b]$ is the largest of the values at the stationary points **and** at a and b .

WATCH OUT In FT Q1(v) both stationary points lie outside $[-1, 1]$, so the answer is at an endpoint and solving $f' = 0$ alone finds nothing.

NEEDED FOR FT Q1(v)

6.9 Constrained optimisation

Use the constraint to eliminate one variable *before* differentiating, or differentiate the constraint to get $\frac{dr}{da}$ and substitute.

WATCH OUT Treating the two variables as independent and setting both derivatives to zero collapses the problem to nothing.

NEEDED FOR PB Q5(b), FT Q14(OR), Exemplar Ch 6 Q31, NCERT Ch 6

6.10 Maximise the square when a root is in the way

A^2 and A have their maximum at the same place when $A > 0$, and A^2 has no square root in it.

NEEDED FOR Exemplar Ch 6 (largest inscribed rectangle), ISC Specimen 2023

7. Integrals

Only three questions in the two papers — the 2024-25 paper has no integration at all — but all three are property-and-identity questions rather than routine ones.

7.1 The half-angle identities

$$1 - \cos x = 2 \sin^2 \frac{x}{2}, \quad 1 + \cos x = 2 \cos^2 \frac{x}{2}, \quad \sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$$

WATCH OUT These turn an apparently hopeless quotient into cosec^2 and $\cot$, which integrate at sight.

NEEDED FOR PB Q1(e), NCERT Ch 7

7.2 $\frac{f'(x)}{f(x)}$ integrates to $\log|f(x)|$

Before any substitution, check whether the numerator is the derivative of the denominator.

NEEDED FOR PB Q1(e), NCERT Ch 7

7.3 Integration by parts, and the cancellation pattern

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

WATCH OUT In PB Q1(e) the term parts produces is exactly the one already waiting to be integrated, and the two cancel. Look for that before grinding on.

NEEDED FOR PB Q1(e)

7.4 The $a - x$ reflection property

$$\int_0^a f(x) dx = \int_0^a f(a - x) dx$$

WATCH OUT On $[0, \frac{\pi}{2}]$ it swaps $\sin$ and $\cos$, and $\tan$ and $\cot$. Adding the two forms usually gives $2I = \text{something trivial}$.

NEEDED FOR PB Q1(f), Exemplar Ch 7 Q30

7.5 Rewrite before you substitute

$\frac{1}{1 + \tan x} = \frac{\cos x}{\cos x + \sin x}$ is what exposes the symmetry; in that form the property works, in the original it does not.

NEEDED FOR PB Q1(f), NCERT Ch 7 Example 6

7.6 Combining $\sqrt{\tan x}$ and $\sqrt{\cot x}$

Put them over a common denominator to get $\frac{\sin x + \cos x}{\sqrt{\sin x \cos x}}$, then use

WATCH OUT $2 \sin x \cos x = 1 - (\sin x - \cos x)^2$ — not $\sin 2x$, which leaves a root you cannot substitute into.

NEEDED FOR PB Q6(a), Exemplar Ch 7 Q43

7.7 Changing the limits with the variable

In a definite integral, either convert the limits when you substitute or convert back before evaluating. Never mix the two.

NEEDED FOR PB Q1(f)

8. Application of Integrals

One question in the two papers, but it needs the whole method — and the first step is not integration at all.

8.1 Identify the curve first

$y^2 = 2x - x^2$ rearranges to $(x - 1)^2 + y^2 = 1$: a **circle** of radius 1 centred at (1, 0).

WATCH OUT Complete the square before anything else. Integrating blindly turns a quarter circle into an intractable integral.

NEEDED FOR PB Q6(b), Exemplar Ch 8 Q16

8.2 Area between two curves

$$A = \int_a^b (y_{\text{upper}} - y_{\text{lower}}) dx$$

WATCH OUT Decide which curve is on top by testing one convenient x between the intersections.

NEEDED FOR PB Q6(b), Exemplar Ch 8 Q16

8.3 Split where the boundary changes

If the upper curve changes, or a modulus changes definition, the integral splits at that point.

NEEDED FOR Exemplar Ch 8 Q18 (triangle), Q23 (modulus)

8.4 Area is not the signed integral

Where the curve dips below the axis the integral is negative. An 'area' answer of zero always means a missing modulus.

NEEDED FOR Exemplar Ch 8 Q17 ($\sin x$ over $[0, 2\pi]$)

8.5 Standard areas worth recognising

- A quarter circle of radius a has area $\frac{\pi a^2}{4}$.
- A sector of angle θ has area $\frac{1}{2}r^2\theta$.
- Integrating along y is often shorter when both curves give x in terms of y .

NEEDED FOR PB Q6(b), Exemplar Ch 8 Q27, Q29

9. Linear Programming

Five questions — and the two long ones are the **same diet problem**, set eight years apart with the grains and prices exchanged. Formulation is most of the marks.

9.1 Formulating

1. Name the variables, with their units.
2. One inequality per resource or requirement.
3. Write the objective, and say whether it is to be maximised or minimised.
4. Add $x, y \geq 0$.

WATCH OUT **Convert the units first.** Costs quoted per kg against contents quoted per gram is exactly where these questions catch people.

NEEDED FOR FT Q22, PB Q13(b), Exemplar Ch 12 Q23

9.2 Half planes, strict and non-strict

A linear inequality is a half plane. A strict one ($<$) excludes its boundary; test the origin to see which side is meant.

NEEDED FOR FT Q19(ii), Exemplar Ch 12

9.3 The corner point theorem

On a **bounded** feasible region the optimum of a linear objective is attained at a corner point. Evaluate Z at each and compare.

NEEDED FOR FT Q19(v), FT Q20, Exemplar Ch 12 Q1, Q10

9.4 Unbounded regions need one more check

If the region is unbounded, a corner value M is the minimum only if the open half plane $ax + by < M$ has **no point in common** with the region.

WATCH OUT Quoting the smallest corner value without this check is an incomplete answer — a minimum need not exist at all.

NEEDED FOR FT Q22, PB Q13(b)

9.5 Ties between two corners

If Z takes the same value at two corner points, the objective line is parallel to that edge and **every** point of the edge is optimal.

NEEDED FOR FT Q19(v), Exemplar Ch 12 Q22 (the bikes)

9.6 Feasible solutions against the optimal solution

Every point of the region is a feasible solution — so there are usually infinitely many. Only the *optimum* is confined to the corners.

WATCH OUT FT Q20 asks for the number of feasible solutions, not the maximum. The answer is 'infinite'.

NEEDED FOR FT Q20

9.7 Not every pair of lines gives a corner

Two constraint lines may meet outside the feasible region, and a constraint may be redundant. Test each intersection against *all* the constraints.

NEEDED FOR Exemplar Ch 12 Q21

Companion to *School Paper Based*. Every concept here is one that a question in that deck actually uses.